

Chap 6. Doob-Meyer type decomposition for rough paths

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April 24th, 2020

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Introduction

Definition (Continuous semi-martingale)

[RY91] A continuous process $(S_t : t \geq 0)$ on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ is a *continuous semi-martingale* if it can be written in

$$S_t = M_t + A_t,$$

where M is a continuous local-martingale and A is a continuous adapted process of finite variation.

Notation.

$\mathcal{M} :=$ (the space of continuous local martingales)

$\mathcal{V} :=$ (the space of continuous adapted process of finite variation)

Proposition (Proposition 6.1. in Textbook)

Assume that $M, \tilde{M} \in \mathcal{M}$, vanishing at zero, and $A, \tilde{A} \in \mathcal{V}$ satisfies

$$M + A \equiv \tilde{M} + \tilde{A}.$$

Then,

$$M \equiv \tilde{M} \quad \text{and} \quad A \equiv \tilde{A}.$$

Furthermore, if $S = M + A \equiv 0$ on some random interval $[0, \tau)$ where τ is a stopping time, then $[M] \equiv 0$ on $[0, \tau)$ and $A \equiv 0$ on $[0, \tau)$.

Corollary (Corollary 6.2. in Textbook)

Let B be a d -dimensional Brownian motion and let $Y, Z, \tilde{Y}, \tilde{Z}$ be continuous stochastic process adapted to the filtration generated by B . Assume that

$$\int_0^\cdot Y dB + \int_0^\cdot Z dt \equiv \int_0^\cdot \tilde{Y} dB + \int_0^\cdot \tilde{Z} dt \quad \text{on } [0, T].$$

Then, $Y \equiv \tilde{Y}$ and $Z \equiv \tilde{Z}$ on $[0, T]$.

Proof. It suffices to consider the case $\tilde{Y} = 0$ and $\tilde{Z} = 0$ and show $Y \equiv 0$. By the previous proposition,

$$\left[\int_0^\cdot Y dB \right]_\cdot \equiv 0 \quad \Rightarrow \quad \left[\sum_{k=1}^d \int_0^\cdot Y_k dB^k \right]_\cdot \equiv 0 \quad \text{on } [0, T].$$

Note the fact that $d[B^k, B^l]_t = \delta^{kl} dt$. Then, it follows from

$$\left[\sum_{k=1}^d \int_0^\cdot Y_k dB^k \right]_\cdot \equiv \sum_{k,l=1}^d \int_0^\cdot Y_k Y_l d[B^k, B^l] = \sum_{k=1}^d \int_0^\cdot Y_k^2 dt$$

that $Y \equiv 0$. □

Question. Suppose

$$\int_0^\cdot Y d\mathbf{X} + \int_0^\cdot Z dt \equiv \int_0^\cdot \tilde{Y} d\mathbf{X} + \int_0^\cdot \tilde{Z} dt$$

holds on $[0, T]$. Then, under what condition do $Y \equiv \tilde{Y}$ and $Z \equiv \tilde{Z}$ holds?

Answer. For all $0 \leq s \leq t \leq T$, one has

$$\int_s^t (Y - \tilde{Y}) d\mathbf{X} = \int_s^t (Z_r - \tilde{Z}_r) dr = O(|t - s|)$$

from the continuity of Z .

For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, one can obtain

$$\int_s^t (Y - \tilde{Y}) d\mathbf{X} = O(|t - s|^{2\alpha}).$$

Recall the Definition 4.6. in Textbook:

$$Y_{s,t} = Y'_s X_{s,t} + O(|t - s|^{2\alpha})$$

and note that

$$I' = Y - \tilde{Y} \quad \text{where} \quad I = \int_0^\cdot (Y - \tilde{Y}) d\mathbf{X}$$

Hence, uniqueness of I' implies $Y \equiv \tilde{Y}$.

- Our question turns to under what condition is the uniqueness of Gubinelli derivative guaranteed?

Roughness and its example

Roughness and its example

Definition (Rough, Definition 6.3. in Textbook)

For fixed $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $s \in [0, T)$, $X \in \mathcal{C}^\alpha([0, T], V)$ is *rough at time s* if the following relation holds for any $v^* \in V^* \setminus \{0\}$:

$$\limsup_{t \rightarrow s+} \frac{|v^*(X_{s,t})|}{|t - s|^{2\alpha}} = \infty.$$

And X is *truly rough* if X is rough on some dense subset of $[0, T]$.

Roughness and its example

Theorem (Theorem 6.6. in Textbook)

With probability one, Brownian motion on $V = \mathbb{R}^d$ is truly rough, relative to any Hölder exponent $\alpha \in [\frac{1}{4}, \frac{1}{2})$.

Proof. It suffices to show that for fixed time s and $\alpha \in [\frac{1}{4}, \frac{1}{2})$,

$$P \left[\limsup_{t \rightarrow s+} \frac{|\varphi(B_{s,t})|}{|t - s|^{2\alpha}} = \infty, \forall \varphi \in V^*, |\varphi| = 1 \right] = 1.$$

We will use the following well-known fact that for all $t \geq 0$,

$$P \left[\limsup_{h \rightarrow 0+} \frac{|B_{t,t+h}|}{\psi(h)} = \sqrt{2} \right] = 1,$$

where $\psi(h) := h^{\frac{1}{2}} (\log \log \frac{1}{h})^{\frac{1}{2}}$.

Roughness and its example

Then, there exists $c > 0$ such that for every fixed unit dual vector $\varphi \in V^*$ and every fixed $s \in [0, T)$,

$$P \left[\limsup_{t \rightarrow s+} \frac{|\varphi(B_{s,t})|}{\psi(t-s)} \geq c \right] = 1.$$

Take $K \subset V^*$ to be dense, countable subset of unit dual vectors. Since K is countable, one has

$$P \left[\limsup_{t \rightarrow s+} \frac{|\varphi(B_{s,t})|}{\psi(t-s)} \geq c, \varphi \in K \right] = 1,$$

and due to the density of K , every unit dual vector $\varphi \in V^*$ is the limit of some $(\varphi_n) \subset K$.

Roughness and its example

Then,

$$c \leq \limsup_{t \rightarrow s+} \frac{|\varphi_n(B_{s,t})|}{\psi(t-s)} \leq \limsup_{t \rightarrow s+} \frac{|\varphi(B_{s,t})|}{\psi(t-s)} + |\varphi_n - \varphi|_{V^*} \limsup_{t \rightarrow s+} \frac{|B_{s,t}|}{\psi(t-s)}$$

almost surely and we let n go to infinity to obtain

$$c \leq \limsup_{t \rightarrow s+} \frac{|\varphi(B_{s,t})|}{\psi(t-s)} \quad \text{almost surely.}$$

Hence, we can take (t_n) converging to s such that

$$\begin{aligned} \frac{|\varphi(B_{s,t_n})|}{|t_n - s|^{2\alpha}} &= \frac{|\varphi(B_{s,t_n})|}{\psi(t_n - s)} \frac{\psi(t_n - s)}{|t_n - s|^{2\alpha}} \\ &\geq \left(c - \frac{1}{n}\right) |t_n - s|^{\frac{1}{2} - 2\alpha} \left(\log \log \frac{1}{t_n - s}\right)^{\frac{1}{2}} \rightarrow \infty \end{aligned}$$

for almost every sample path. □

Doob-Meyer for rough paths

Proposition (Proposition 6.4. in Textbook)

(Uniqueness of Y') Let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$, $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, so that the rough integral $\int Y d\mathbf{X}$ exists. Suppose X is rough at some time s . Then,

$$Y_{s,t} = O(|t - s|^{2\alpha}) \quad \text{as } t \rightarrow s+ \quad \implies \quad Y'_s = 0.$$

As a consequence, if X is truly rough and $(Y, \tilde{Y}') \in \mathcal{D}_X^{2\alpha}$ is another controlled rough path, then $Y' \equiv \tilde{Y}'$.

Proof. From the definition of (Y, Y') , one have

$$\frac{Y'_s X_{s,t}}{|t - s|^{2\alpha}} = \frac{Y_{s,t}}{|t - s|^{2\alpha}} + O(1) = O(1), \quad s < t < s + \epsilon.$$

For every $w^* \in \overline{W}^*$, the map

$$V \ni v \mapsto w^*(Y'_s v)$$

defines an element $v^* \in V^*$ so that

$$\frac{|v^*(X_{s,t})|}{|t-s|^{2\alpha}} = \left| w^* \left(\frac{Y'_s X_{s,t}}{|t-s|^{2\alpha}} \right) \right| = O(1) \quad \text{as } t \rightarrow s+.$$

Since X is rough at s , we have $v^* = 0$, which implies that $Y'_s = 0$. $\square$

Doob-Meyer for rough paths

Theorem (Theorem 6.5. in Textbook)

(Doob-Meyer for rough paths) Suppose that X is rough at some time $s \in [0, T)$ and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}$. Then,

$$\int_s^t Y d\mathbf{X} = O(|t - s|^{2\alpha}) \quad \text{as } t \rightarrow s+ \implies Y_s = 0.$$

As a consequence, if X is truly rough, $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_X^{2\alpha}$ and $Z, \tilde{Z} \in \mathcal{C}([0, T], W)$, then the identity

$$\int_0^\cdot Y d\mathbf{X} + \int_0^\cdot Z dt \equiv \int_0^\cdot \tilde{Y} d\mathbf{X} + \int_0^\cdot \tilde{Z} dt$$

on $[0, T]$ implies that $(Y, Y') \equiv (\tilde{Y}, \tilde{Y}')$ and $Z \equiv \tilde{Z}$ on $[0, T]$.

Quantitative version of roughness

Quantitative version of roughness

Definition (θ -Hölder rough, Definition 6.7. in Textbook)

A path $X : [0, T] \rightarrow V$ with values in a Banach space V is said to be θ -Hölder rough for $\theta \in (0, 1)$, on scale (smaller than) $\epsilon_0 > 0$, if there exists a constant $L := L_\theta(X) := L(\theta, \epsilon_0, T; X) > 0$ such that for every $\varphi \in V^*$, $s \in [0, T]$ and $\epsilon \in (0, \epsilon_0]$ there exists $t \in [0, T]$ such that

$$|t - s| < \epsilon \quad \text{and} \quad |\varphi(X_{s,t})| \geq L_\theta(X) \epsilon^\theta |\varphi|.$$

The largest such value of L is called the *modulus of θ -Hölder roughness* of X .

Remark. Every element in $\mathcal{C}^\alpha$ which is θ -Hölder rough for $\theta < 2\alpha$ is truly rough.

Proposition (Proposition 6.8. in Textbook)

Let $(X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$ be such that X is θ -Hölder rough for some $\theta \in (0, 1]$. Then, for every controlled rough path $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \overline{W})$ one has,

$$L\epsilon^\theta \|Y'\|_\infty \leq \text{osc}(Y, \epsilon) + \|R^Y\|_{2\alpha} \epsilon^{2\alpha}.$$

As a consequence, if $\theta < 2\alpha$, Y' is uniquely determined from Y .

Remark. This Proposition is quantitative version of Proposition 6.4. in Textbook.

Quantitative version of roughness

Proof. Fix $s \in [0, T]$ and $\epsilon \in (0, \epsilon_0]$. From the definition of Gubinelli derivative, one has

$$\sup_{|t-s| \leq \epsilon} |Y'_s X_{s,t}| \leq \sup_{|t-s| \leq \epsilon} (|Y_{s,t}| + |R_{s,t}^Y|) \leq \text{osc}(Y, \epsilon) + \|R^Y\|_{2\alpha} \epsilon^{2\alpha}.$$

Since X is θ -Hölder rough, for unit $\varphi \in W^*$, there exists $t_0 = t_0(\varphi)$ with $|t_0 - s| < \epsilon$ such that

$$|\varphi(Y'_s X_{s,t_0})| = |((Y'_s)^* \varphi)(X_{s,t_0})| > L \epsilon^\theta |((Y'_s)^* \varphi)|.$$

Hence, from the arbitrariness of φ and s , one can obtain

$$L \epsilon^\theta |Y'| \leq \text{osc}(Y, \epsilon) + \|R^Y\|_{2\alpha} \epsilon^{2\alpha}.$$

□

Quantitative version of roughness

Theorem (Theorem 6.10. in Textbook)

Let $(X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$ be such that X is θ -Hölder rough with $\theta < 2\alpha$. Let $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \overline{W})$ and $Z \in \mathcal{C}^\alpha([0, T], W)$ and set

$$I_t := \int_0^t Y_s d\mathbf{X}_s + \int_0^t Z_s ds.$$

Then, there exist constants $r > 0$ and $q > 0$ such that

$$\|Y\|_\infty + \|Z\|_\infty \leq M \mathcal{R}^q \|I\|_\infty^r,$$

where constant M only depends on α, θ and T and

$$\mathcal{R} := 1 + L_\theta(X)^{-1} + |||X|||_\alpha + \|Y, Y'\|_{X; 2\alpha} + |Y_0| + |Y'_0| + \|Z\|_\alpha + |Z_0|.$$

Remark. This Theorem is quantitative version of Theorem 6.5. in Textbook.

Summary

1. X is rough at time $s \iff \limsup_{t \rightarrow s+} \frac{|v^*(X_{s,t})|}{|t-s|^{2\alpha}} = \infty, \forall v^* \in V^*$
2. Brownian motion is truly rough.
3. Roughness of X implies the uniqueness of Gubinelli derivative and therefore, the uniqueness of Doob-Meyer type decomposition.

THANK YOU FOR LISTENING